MTH401 FINAL TERM SOLVED PAPER

BY AQUALEO

(VISIT VURANK FOR MORE)

_____

Find order of homogenous equation obtained from non homogenous differential equation: $y¢¢+ 4y¢+3y = 4x^2 +5?$**? (2 MARKS)**

Find the eigenvalues of the following system

Ê3 -9ˆ

$X¢$ Á ˜ X

Ë4 -3¯

Solution:

Ê3 -9ˆ

X' ÁË 4 -3˜¯ X

A ÊÁË 43 --93ˆ ˜¯ for

eigen values, $A-lI = 0$

3-l -9
4 -3-l

$(3-l)(-3-l)+36 = 0$

$3(-3-l)-l(-3-l)+ 36 = 0$

$-9\ -3\lambda+3\lambda+\lambda^2+36\ =\ 0\ -9$

$+\lambda^2+36 = 0\ \lambda^2 + 27 = 0$

$\lambda = 27i$ and $-27i$ are the two complex eigen values

Q:3

What is Chemical reaction first order equation? (2) **Page no 100**

Answer:

$$\frac{dX}{dt} = k\,X$$

$k<0$ because X is decreasing.

Q:4

What is charachteristic equation? **Page no 379**

Answer:

$$\det(A-\lambda I)=0$$

This equation is called the characteristic equation of the matrix A.

Q:5

Can we extend power series?

AnsweR: **Page no 268**

I answered in yes and then wrote the extended form of power series.

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots.$$

Q:6 Page no 371

Find the derivative and the integral of the following matrix

$$X(t)=\begin{pmatrix}\sin 2t\\ e^{3t}\\ 8t-1\end{pmatrix}$$

Solution:

The derivative and integral of the given matrix are, respectively, given by

$$X'(t)=\begin{pmatrix}\frac{d}{dt}(\sin 2t)\\ \frac{d}{dt}(e^{3t})\\ \frac{d}{dt}(8t-1)\end{pmatrix}=\begin{pmatrix}2\cos 2t\\ 3e^{3t}\\ 8\end{pmatrix}$$

Q:7

Write system of equation in matrix form?

Solution: Page no 387

$\frac{dx}{dt} = -3x + 4y - 9z$

$\frac{dy}{dt} = 6x - y$

$\frac{dz}{dt} = 10x + 4y + 3z$

Solution :

È*dx* ˘

Í*dt* ˙ = È -3 4 -9˘ È*x* ˘

Í*dy* ˙ =Í6 -1 0 ˙ Í*y* ˙

Í*dt* ˙ Í ˙ Í ˙ ÍÍ *dz* ˙ ˙

ÍÎ10 4 3˙ Í˚Î*z* ˙ ˚

Í*dt* ˙ ˚

Q:8 Page no 98

Dudce special case of logistic equation (epidemic spread)? (5)

The natural assumption is that the rate $\frac{dx}{dt}$ of spread of disease is proportional to the number $x(t)$ of the infected people and number $y(t)$ of people not infected people. Then

$$\frac{dx}{dt} = kxy$$

Since $x + y = n + 1$

Therefore, we have the following initial value problem

$$\frac{dx}{dt} = kx(n+1-x), \qquad x(0) = 1$$

The last equation is a **special case of the logistic equation** and has also been used for the **spread of information** and the **impact of advertising** in centers of population.

Q:9

Find order of homogenous equation obtained from non homogenous differential equation:

$y¢¢ + 4y¢ + 3y = 4x^2 + 5$**? (2 MARKS)**

Q:10:

$$y¢¢ + y = 0$$

Find a series solution for the differential equation about $x_0 = 0$ such that

Find condition of cofficent for a_{n+2} & a_n (c_{n+2} & c_n)?

Q:11

Which series is identically zero? Page no 273

Answer:

Series that are Identically Zero

If for all real numbers x in the interval of convergence, a power series is identically zero i.e.

$$\sum_{n=o}^{\infty} c_n (x-a)^n = 0, \qquad R>0$$

Then all the coefficients in the power series are zero. Thus we can write

$$c_n = 0, \qquad \forall\ n = 0, 1, 2, \ldots$$

Q:12

A ÈÍÎ 32 --189 ˘ ˙ ˚

eigenvalues ?

Eigenvectors ?

Note: I am not going to solve this question solve it by your self by consulting two examples below.

========== First Paper End ==========

Q1: Find Coefficient of metrix:

$$\frac{dx}{dt} = -3x - 2y$$

$$\frac{dy}{dt} = 5x + 7y$$

Solution:

Cofficent of matrix =

$$A = \begin{bmatrix} -3 & -2 \\ 5 & 7 \end{bmatrix}$$

Q2: Eigen Values of metrics.

A=ÈíÎ 02 33˘ · °

Consider the question below:

Ê3 -9ˆ

A ÁË 4 -3˜¯

for eigen values, $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & -9 \\ 4 & -3-\lambda \end{vmatrix} = 0$$

$(3-\lambda)(-3-\lambda)+36 = 0$

$3(-3-\lambda)-\lambda(-3-\lambda)+36 = 0$

$-9-3\lambda+3\lambda+\lambda^2+36 = 0$ -

$9+\lambda^2+36 = 0 \quad \lambda^2 + 27 = 0$

$\lambda = 3\sqrt{3}i$ and $-3\sqrt{3}i$ are the two complex eigen values **This**

question is similar to above.

Q3: whether or not a singular points have real number if not then give some examples?

Answer: **Page no 284**

(b) The singular points need not be real numbers.

The equation $(x^2+1)y''+2xy'+6y=0$ has the singular points at the solutions of $x^2+1=0$, namely, $x=\pm i$.

Q4: Solve the differential equation. $\frac{1}{y}\frac{dy}{dx}$ 1

Solution:

$$\frac{1}{y}\frac{dy}{dx} \quad 1$$

$$\frac{dy}{y} \quad (1)dx$$

$$\int \frac{dy}{y}$$

$$\int (1)dx \ln y$$

$$= x + c \; y$$

$$e^{x+c}$$

Q5: complementary solution of DE

$$y'' - 4y' + 4y = 2e^{2x}$$

Solution: **Page no 182**

$$y'' - 2y' + y = 0$$

The auxiliary equation is

$$m^2 - 2m + 1 = 0 \Rightarrow (m-1)^2 = 0 \Rightarrow m = 1, 1$$

The complementary function for the given equation is

$$y_c = c_1 e^x + c_2 x e^x$$

Q6: state the Bessel's function of first kind of order ½ and -1/2.

Solution: **Page no 313**

$$J_v(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)\Gamma(1+v+n)} \left(\frac{x}{2}\right)^{2n+v} \quad (6)$$

Also for the second exponent $r_2 = -v$, we have

$$J_{-v}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)\Gamma(1-v+n)} \left(\frac{x}{2}\right)^{2n-v} \quad (7)$$

Only put the value of ½ in $J_v(x)$ and - ½ in $J_{-v}(x)$ at the places of v.

Q7: Define the derivative of

Èe_{2t} ˘

A (t) = ÍÍt_2 · ·

Í8 · Î

˚

Answer: **Repeated**

Q8: Find the egien values of

È ˘

Í1 -1 ·
Í ·
A Í ·

Í 4 -1·
Í__ __· ÍÎ9

3 · °

Solution:

Consider the question below.

Ê3 -9ˆ

A ÁË 4 -3˜¯

for eigen values, $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & -9 \\ 4 & -3-\lambda \end{vmatrix} = 0$$

$(3-l)(-3-l)+36 = 0$

$3(-3-l)-l(-3-l)+36 = 0$

$-9-3l+3l+l^2+36 = 0$ -

$9+l^2+36 = 0\ l^2+27 = 0$

$l = 2\sqrt{i}$ and $-27i$ are the two complex eigen values

Q9: bht lamba tha mery sy note ni hoa time thora tha is lia ☹

Q10: Find the auxiliary solution of $x^t = 3x - y - 1$ **and** $y^t = y+x-4e^t$

Consult page no 141

Q11: Write down the system of differential equations (5 marks) dx

dy

$\frac{}{dt} = 6x+y+6t$, $\frac{}{dt} = 4x+3y-10t+4$

In form of $X' = AX + F(t)$

Solution:

$$X' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} X + \begin{pmatrix} 6t \\ -10t + 4 \end{pmatrix}$$

========== PAST PAPERS==========

Q: An electronic component of an electronic circuit that has the ability to store charge and opposes any change of voltage in the circuit is called

Inductor

Resistor

Capacitor

None of them

Q: If A_o is initial value and T denotes the half-life of the radioactive substance than

$T \quad \frac{1}{2A}$

$\frac{dA}{dt} \quad KA$

$A(T) \quad \frac{A_0}{2}$

None of the above

Q: integrating factor of the given equation $x\cos x \frac{dy}{dx} + y(x\sin x + \cos x)$ is

Xsecx

Cosx

Cotx

Xsinx

Q: Operator method is the method of the solution of a system of linear homogeneous or linear non-homogeneous differential equations which is based on the process of systematic elimination of the

Dependent variables

Independent variable
Choice variable
None of them

Q: If E (t) =0, R =0 Electric vibration of the circuit is called_____

Free damped oscillation

Un- damped oscillation

Over damped oscillation

None of the given

Q: Eigen value of a matrix Ê ÁË3 1 7 4 ˆ ~ –

5 , 5

10 , 5

25 , 5

None

Q: Eigen value of a matrix Ê ÁË 1 1 1 1 ˆ ~ –

2 , 0 1

, 1 1 ,

2

None

A ÊÁË31 47ˆ ~–

Q: For Eigen values l 5,5 of a matrix ,there exists..... Eigen vectors.

infinite

one
two
three

Q: If a matrix has 1 row and 3columns then the given matrix is called_____ Column matrix

Row matrix

Rectangular matrix None

$$\frac{dy}{dx} \quad \frac{x+y}{x}$$

Q: The general solution of differential equation .is given by

$\frac{y}{x}$

$\frac{y}{x}$

$\frac{x}{y}$

$ecx\ ecy$

ecx

$e^{\frac{x}{y}}cx$

Q: The integrating factor of the D.E $\frac{dy}{dx} + y\ln y = ye^x$ is

$e_x\ e_y$

$\frac{1}{e_x}$

$\frac{x}{e_y}$

Q: For the equation of free damped motion $\frac{dx}{dt}^2 + 2l\frac{dx}{dt} + w^2 x = 0$ the roots are m_1 $= -l + \sqrt{l^2 + w^2}$ & $m_1 = -l - \sqrt{l^2 + w^2}$ if $l^2 - w^2 > 0$ Then the equations said to be:

Under damped
Over damped

Critically damped
None of them

Q: For the system of differential equations $\frac{dy}{dt} = 2x, \frac{dx}{dt} = 3y$ the independent variable is

(**Are)**

X,t

Y,t

X,y

t

Q: For the system of differential equations $\frac{dy}{dt} = 2x, \frac{dx}{dt} = 3y$ the dependent variable is

(**Are)**

X,t

Y,t

X,y

t

Ê4-l 1 0 ˆ

Q:Á Á 0 4-l 1 ˜˜= 0 gives

ÁË 0 0 4-l˜¯

l 4 of multiplicity of 1 l 4

of multiplicity of 2 l 4 of

multiplicity of 3

None of the given.

Q: wronksin of x, x^2 is

x^2

X

O

None of the above

a) Matrix A nd value of lembda was given to find the eigen vector? 3 marks.

Answer: (This question is solved by Shining Star as original question was missing so I put it here for reference.)

Ê-3 1 ˆ

ÁË 2 -4˜¯

A= , corresponding Eigen valuel=-2**.**

Ê-3-(-2) 10ˆ
Á˜

Ë 2 -4-(-2)0¯

Ê-1 1 0ˆ

Á ~

Ë 2 -2 0¯

Add two times row 1 in row 2

Ê-1 1 0ˆ
Á ~
Ë 0 0 0¯

$-k_1 + k_2 = 0$

$k_1 \quad k_2$

Choosing $k_2 = 1$, we get $k_1 = 1$

Ê1ˆ

therefore, eigen vector is V ÁË 1˜ ¯

b) X'=AX was given to find the eigenvalue and Eigen vector? 5 marks. (This question is solved by Shining Star as original question was missing so I put it here for reference.)

For eigen values consut this question and for eigen vector look at the above.

Ê3 -9ˆ

$X¢$ ÁË 4 -3˜¯ X

Solution:

Ê3 -9ˆ

X' ÁË 4 -3˜¯ X

A ÊÁË 43 --93ˆ ˜¯ for

eigen values, $|A-I| = 0$

$$\begin{vmatrix} 3-\lambda & -9 \\ 4 & -3-\lambda \end{vmatrix}$$

$$(3-\lambda)(-3-\lambda)+36 = 0$$

$$3(-3-\lambda)-\lambda(-3-\lambda)+ 36 = 0$$

$$-9 \ -3\lambda+3\lambda+\lambda^2 +36 \ = \ 0 \ -9$$

$$+\lambda^2 +36 = 0 \ \lambda^2 + 27 = 0$$

$\lambda = 2\sqrt{7}i$ and $-27i$ are the two complex eigen values

c) Solve DE dy-7dx=0 for initial value f(0)=1? 5 marks.

Answer:

$dy - 7dx = 0$

$dy \quad 7dx$

$$\acute{U}_{dy}$$

$$\acute{U}_{7d}$$

$x\,y = 7(x$

$)+c$

$f(0) \quad 1$

$f(0) = 7(0)+c$

$f(0) = 0+c$

$1\ C$

$y = 7x + 1$

d) Find the general solution of 4x^2 y '' + 4xy' (4x^2-25)y=0 (it is the Bessel's Equation a nd same question is given in exercise pg 314 of our handouts)? 5 marks

Answer:

Bessel's differential equation is

$$x^2 y'' + xy' + \left(x^2 - v^2\right) y = 0$$

Example 1

Find the general solution of the equation

$$x^2 y'' + xy' + \left(x^2 - \frac{1}{4}\right) y = 0 \text{ on } (0, \infty)$$

Solution

The Bessel differential equation is

$$x^2 y'' + xy' + \left(x^2 - v^2\right) y = 0 \qquad (1)$$

$$x^2 y'' + xy' + \left(x^2 - \frac{1}{4}\right) y = 0 \qquad (2)$$

Comparing (1) and (2), we get $v^2 = \frac{1}{4}$, therefore $v = \pm\frac{1}{2}$

So general solution of (1) is $y = C_1 \mathrm{J}_{1/2}(x) + C_2 \mathrm{J}_{-1/2}(x)$

Answer:

e) When a function is said to be analytic at any point? 2 marks

Answer:

A function is said to be analytic at point if the function can be represented by power series in (x−a) with a positive radius of convergence.

f) What is the ratio test? (its on pg 264 of our handouts) 5 marks

To determine for which values of x a power series is convergent, one can often use the Ratio Test. The Ratio test states that if

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} c_n (x-a)^n$$

is a power series and

$$\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right| |\, x-a \,| = L$$

Then:

- The power series converges absolutely for those values of x for which $L<1$.
- The power series diverges for those values of $x$for which $L>1$ or $L = \infty$.
- The test is inconclusive for those values of x for which $L=1$.

g) What is the formula for radius of convergence? (Its on pg 265 of our hndouts)2 marks Answer:

$$R = \lim_{n\to\infty} \left| \frac{c_n}{c_{n+1}} \right|$$

h) Write system of linear differential equations for two variables x and y?

(its on pg 333 of our handouts).2 marks

i) write any 3 D.Es of order 2? 3 marks Page no 207 Answer:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$$

j) D.E was given to convert in normal form? 3 marks Answer:

Reduce the third-order equation

$$2y''' = -y - 4y' + 6y'' + \sin t$$

or $$2y''' - 6y'' + 4y' + y = \sin t$$

to the normal form.

Solution: Write the differential equation as

$$y''' = -\frac{1}{2}y - 2y' + 3y'' + \frac{1}{2}\sin t$$

Now introduce the variables

$$y = x_1, y' = x_2, y'' = x_3.$$

Then

$$x_1' = y' = x_2$$
$$x_2' = y'' = x_3$$
$$x_3' = y''$$

Hence, we can write the given differential equation in the linear normal form

$$x_1' = x_2$$
$$x_2' = x_3$$
$$x_3' = -\frac{1}{2}x_1 - 2x_2 + 3x_3 + \frac{1}{2}\sin t$$

k) Any example of boundary value problem? 2 marks

Consider the function

$$y = 3x^2 - 6x + 3$$

We can prove that this function is a solution of the boundary-value problem

$$x^2 y'' - 2xy' + 2y = 6,$$
$$y(1) = 0, \quad y(2) = 3$$

Since $$\frac{dy}{dx} = 6x - 6, \quad \frac{d^2y}{dx^2} = 6$$

Therefore $$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 6x^2 - 12x^2 + 12x + 6x^2 - 12x + 6 = 6$$

Also $$y(1) = 3 - 6 + 3 = 0, \quad y(2) = 12 - 12 + 3 = 3$$

Therefore, the function 'y' satisfies both the differential equation and the boundary conditions. Hence y is a solution of the boundary value problem.

Note: Power series sy ziada NHI tha. Lecture 35 to 45 pr ziada emphsis tha

Q No.2 ----------5 marks:

Write annihilator operator for x+3xe^ (6x) e ki power 6 xs

$$g(x) = 4e^{2x} - 6xe^{2x}$$

$$(D-2)^2\left(4e^{2x} - 6xe^{2x}\right) = (D^2 - 4D + 4)(4e^{2x}) - (D^2 - 4D + 4)(6xe^{2x})$$

or $$(D-2)^2\left(4e^{2x} - 6xe^{2x}\right) = 32e^{2x} - 32e^{2x} + 48xe^{2x} - 48xe^{2x} + 24e^{2x} - 24e^{2x}$$

or $$(D-2)^2\left(4e^{2x} - 6xe^{2x}\right) = 0$$

Therefore, the annihilator operator of the function g is given by

$$L = (D-2)^2$$

We notice that in this case $\alpha = 2 = n$.

Q No.3 ----------3 marks:

Write the solution of simple harmonic motion in alternative simpler form x(t)=c1coswt+c2sinwt from lec 22 page 199

Answer:

Q No.4 ----------2 marks: Define

general linear DE of nth order

Answer:

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

Define elementary row operation.

Answer:

Addition or multiplication of two rows.

The elementary row operations consist of the following three operations

- Multiply a row by a non-zero constant.
- Interchange any row with another row.
- Add a non-zero constant multiple of one row to another row.

Eigenvalue of multiplicity m 3

Answer:

Suppose that m is a positive integer and $(\lambda - \lambda_1)^m$ is a factor of the characteristic equation

$$\det(A - \lambda I) = 0$$

Further, suppose that $(\lambda - \lambda_1)^{m+1}$ is not a factor of the characteristic equation. Then the number λ_1 is said to be an eigenvalue of the coefficient matrix of multiplicity m.

Fundamental of matrix 3

Answer:

Suppose that the a fundamental set of n solution vectors of a homogeneous system $X' = AX$, on an interval I, consists of the vectors

$$X_1 = \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix}, X_2 = \begin{pmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{pmatrix}, \ldots, X_n = \begin{pmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{pmatrix}$$

Then a fundamental matrix of the system on the interval I is given by

$$\phi(t) = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{pmatrix}$$

What is determinnant? How to find it.

Determinant of a Matrix

Associated with every square matrix A of constants, there is a number called the determinant of the matrix, which is denoted by det(A) or $|A|$

Write equation in matrix form.

Find general solution..... 5marks..

Forbenius Theorem.....

$$y = (x - x_0)^r \sum_{n=0}^{\infty} c_n (x - x_0)^n = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$$

5 marks

Super position method for vectors

Answer:

$$y = C_1 y_1(x) + C_2 y_2(x)$$

Explain convergence and infinty condition of a infintye sereies.

- If we choose a specified value of the variable x then the power series becomes an infinite series of constants. If, for the given x, the sum of terms of the power series equals a finite real number, then the series is said to be convergent at x.

What does these symbols mean?

Symbol	Meaning
R_{ij}	Interchange the rows i and j.
cR_i	Multiply the ith row by a nonzero constant c.
$cR_i + R_j$	Multiply the ith row by c and then add to the jth row.

$$\frac{dy}{dt} = x, \frac{dx}{dt} = y$$

Q2. Solve the system of differential equations by systematic elimination.

Solution:

$$\frac{dy}{dt} = x \Rightarrow Dy - x = 0 \text{.....}(i)$$

$$\frac{dx}{dt} = y \Rightarrow -Dx + y = 0 \text{.....}(ii)$$

Operate (ii) by D, we get

$$-Dy + D^2x = 0 \text{.....}(iii)$$

Add (i) and (iii), we get

$$Dy - x = 0$$

$$-Dy + D^2x = 0$$

$D^2x - x = 0$

$(D^2 - 1)x = 0$

Auxiliary equation is $m^2 - 1 = 0$ $m = \pm 1$

$x(t)$ $c_1e_t + c_2e_{-t}$

Put this in (i), we get

$Dy - \text{È}\hat{\text{I}}c_1e_t + c_2e_{-t}\text{˘}\,^\circ = 0$

$Dy = c_1e_t + c_2e_t$

Integrate both sides, we get

$y(t) = c_1e^t - c_2e^{-t}$

$y¢¢ + y = 0$

Q3. Find a series solution for the differential equation about $x_0 = 0$ such that

.

$a_{n+2} = -\dfrac{a_n}{(n+2)(n+1)}$ $n = 0, 1, 2, \ldots$ $y(x)$ $\hat{\text{A}}a_n x^n$ $n\,0$

and

Solution:

$n+2$ _____ a_n ;

$n = 0, 1, 2, \ldots$ a $=-$

$(n+2)(n+1)$

_____ a^0 a_0

For n 0, a

$= \quad {}_2 = - = -$

$$(0+2)(0+1) \qquad 2a^1$$

$$a^{\underline{1}}$$

For $n = 1$, $a_3 = -_____ = -$

$$(1+2)(1+1)6$$

For $n = 2$, $a_4 = -_____(2+2a)^2(2+1) = -12a^{\underline{2}} = -12__1 \quad \hat{E}_{\acute{A}\ddot{E}} - a_{2^{\underline{0}}\tilde{}\hat{}-} = __24\, a_0$

For $n = 3$, $a_5 = -\overline{_____(3+2)(3+1)} = -\underline{20} = -\,20\acute{A}\ddot{E} - 6\,\tilde{}\bar{} = 120\,___y(x) = a_0$

$$\frac{a_3}{} \quad \frac{a_3}{} \quad 1\,\hat{E}\, a_{1\underline{}}\hat{}\, a_1$$

$+ a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \ldots$

$$y(x) = a_0 + a_1x + \hat{E}\acute{A} - a2_{\underline{0}}\,\tilde{}\hat{}\bar{}\,x_2 + \acute{A}\hat{E}\ddot{E} - a6_{\underline{1}}\,\hat{}\tilde{}\bar{}\,x_3 + \acute{A}\ddot{E}\hat{E}__24a_0\,\hat{}\tilde{}\bar{}\,x_4$$
$$+ \hat{E}\acute{A}\ddot{E}120___a_1\,\tilde{}\bar{}\,\hat{}\,x_5 + \ldots\,\ddot{E}$$

$$y(x) = a_0\,\hat{E}\acute{A}1 - \overline{12}\,x_2 + \overline{24}\,1\,x_4 + \ldots\tilde{}\bar{}\hat{} + a_1\bar{\hat{E}}\acute{A}\ddot{E}x - \overline{16}\,x_3 + 1201 \quad x_5 + \ldots\bar{}\,\hat{}\tilde{}$$
$$\ddot{E}$$

$$X = \frac{4}{3}Cos3t - \frac{5}{3}Sin3t$$

$$X = ASin.(wt + f)$$

Q4. Write solution in the form .

$$A = \sqrt{\hat{E}\frac{4}{3}\hat{}\tilde{}^2 + \hat{E}-\frac{5}{3}\hat{}\tilde{}^2} = \frac{\sqrt{41}}{3}$$

$$f = \tan^{-1}\hat{E}\frac{4/3}{-5/3}\hat{}\tilde{} = 0.6747 \text{ radians}$$

$$x(t) = \frac{\sqrt{41}}{3}\sin(3t + 0.6747)$$

If the equation

$$M(x,y)dx + N(x,y)dy = 0$$

Q5. is not exact,

Case 1:

When $\exists$ an integrating factor $u(y)$, a function of y only. This happens if the expression

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

is a function of y **Case2:**

If the given equation is homogeneous and

$$xM + yN \neq 0$$

Then find the integrating factor in both cases.

Solution:

$$u \quad \frac{1}{xM + yN}$$

Q8. Under which conditions linear independence of the solutions for the differential equation $y¢¢+P(x)y¢+Q(x)y = 0$ .....(1) **is guaranteed?**

Solution:

Linear independence is guaranteed in case when the Wronskian of the two solutions is not equal to zero.

Q10. When Frobenius' Theorem is used in Differential

$$a_2(x)y¢¢+a_1(x)y¢+ a_0(x)y = 0$$

Equation ?

Solution:

When we have a regular singular point x= x0, then we can find at least one series solution of the form $y=\hat{A}c_n(x-x_0)^{n+r}$, where r is the constant that we will determine after solving the differential equation.

$n\,0$

Q12. Define Legendre's polynomial of degree n Solution:

Legendre polynomial is an nth degree polynomial and it is given by the formula

$$P_n(x) = \frac{1}{n!\,2^n}\frac{d^n}{dx^n}(x^2-1)^n$$

Q13. What is the ordinary differential equation and give an example?

Solution:

A differential equation which only includes ordinary derivatives is known as ordinary differential equation. Some examples of ordinary differential equations include:

$$\frac{dy}{dx} = x^2 + y$$

$$\frac{dy}{dx} = (x^2-1)(y^2+1)$$

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 3y = 0$$

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